

The Wrong Test for the Right Question: Geometric-Null Binomial Proximity Testing for Longitude Quantization in Monument Distributions

Seth Tenenbaum*
Independent Scholar, USA

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Abstract

Claims that ancient or sacred monuments align with longitude grids have appeared repeatedly in the archaeoastronomical literature and been repeatedly criticised — but almost always on grounds of investigator-assembled site lists rather than statistical framework. This paper identifies a more fundamental problem: the Rayleigh periodogram, the field’s standard circular-statistical tool, is structurally incapable of testing the hypothesis it is applied to. The hypothesis “sites cluster near specific harmonic longitude lines” is a proximity question. The Rayleigh statistic answers a different question: whether all site phases share a single preferred direction. A true grid-placement signal distributes sites near *multiple* harmonic lines simultaneously, producing *low* Rayleigh R , not high. We demonstrate this empirically: a period sweep of the UNESCO context-validated dome/stupa corpus ($N = 83$) shows Rayleigh significance at arbitrary periods (2.4, 3.5, 3.6) with no corresponding harmonic proximity enrichment, while the only period with genuine proximity enrichment (3.0, $p_{\text{binom}} = 0.0098$) is Rayleigh-non-significant. The correct test is a one-sided binomial on site counts within a threshold distance of any harmonic, against the geometric null $p_0 = 2d/T$. We apply this framework to three independently assembled hemispherical monument corpora — UNESCO World Heritage dome/stupa sites ($N = 83$ context-validated), Wikidata Q180987 stupas ($N = 229$), and a crowd-sourced OpenStreetMap stupa layer ($N = 259$) — as a worked demonstration. All three show directional excess over the geometric null at the 3 / $\sim 35^\circ\text{E}$ candidate frame; their independent binomial p-values combine via Fisher’s method to $p = < 0.001$ at the broadest tier. This result is offered as illustration, not proof of a historical grid: corpus-frame conditioning, geographic dependence, and the mechanism gap are fully acknowledged. The methodological contribution — the dissociation between Rayleigh periodicity and harmonic proximity, and the geometric-null binomial framework that correctly tests the latter — applies to any claim of this type regardless of context.

*ORCID: 0009-0008-5797-2498. Correspondence: seth@fourthtemple.com

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1 Introduction

Do monuments cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, usually motivated by observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. It has also been repeatedly attacked — but almost always on grounds of investigator-assembled site lists, insufficient correction for multiple comparisons, or lack of independent replication (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). These are legitimate criticisms, but they leave a more fundamental problem unaddressed: the standard statistical tool applied to this class of question is structurally incapable of testing it.

The tool in question is the Rayleigh statistic R . R measures the degree to which a set of phases concentrates toward a single preferred direction on the unit circle — it detects unimodal directional concentration. Prior studies have applied Rayleigh tests to longitude distributions and interpreted a significant R at some period T as evidence that sites are placed near the harmonics of T (Thom, 1967; Magli, 2013, 2016).

This interpretation is wrong. A grid-placement hypothesis predicts that sites fall near *multiple* harmonic lines at spacing T — near longitudes $\lambda_0, \lambda_0 + T, \lambda_0 + 2T$, and so on. That distributes phases near $0, 2\pi/K, 4\pi/K, \dots$ on the unit circle, which is precisely the pattern that produces a *low* R , not a high one. Rayleigh significance at period T is evidence that all sites share one phase — a single preferred longitude modulo T — which is neither what grid-placement predicts nor what the hypothesis requires. The two statistics address orthogonal questions.

The correct test for “sites cluster near harmonics” is straightforward. Under a uniform longitude distribution, the fraction of sites falling within $\pm d$ of any harmonic of period T is exactly $p_0 = 2d/T$, regardless of geographic clustering. This is the *geometric null*. The appropriate statistic is a one-sided binomial test on the count of sites within threshold d of any harmonic, against expected $N \times p_0$. When multiple independently assembled corpora are available, their binomial p-values combine via Fisher’s method to give a joint estimate not inflated by within-corpus geographic structure, because the corpora’s geographic clusters are non-overlapping.

This paper makes the dissociation explicit, demonstrates it empirically, and proposes the geometric-null binomial framework as the correct inferential tool for this class of hypothesis. We illustrate the framework using three independently assembled corpora of hemispherical monuments as a worked example at a candidate $3 / \sim 35^\circ\text{E}$ frame. The empirical result is offered as illustration rather than proof: we make no claim for a historical grid, intentional placement, or unique origin.

2 Why the Rayleigh Test Fails for Harmonic Proximity

2.1 The Structural Mismatch

Let $\lambda_1, \dots, \lambda_N$ be site longitudes and T a candidate grid period. The Rayleigh statistic maps each longitude to a phase $\theta_i = 2\pi(\lambda_i \bmod T)/T$ and computes

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|.$$

R is large when all θ_i concentrate near the same value: when sites cluster near a *single* longitude modulo T . R is small when θ_i are dispersed, including the case where they concentrate at *multiple* equally spaced values.

A grid-placement hypothesis predicts the latter. If a set of harmonic longitudes $\{\lambda_0 + kT : k = 0, 1, 2, \dots\}$ attracts monument placement, sites will cluster near each harmonic separately. For any geographically spread corpus with sites near several different harmonics, R will be small. High Rayleigh R at period T would instead indicate that all sites fall near the *same* harmonic — a much stronger and less plausible claim.

A significant Rayleigh R at an arbitrary period T in a geographically clustered corpus can arise whenever one geographic cluster dominates the phase distribution — not because sites are near harmonics, but because they are near one longitude and that longitude produces a peaked distribution at the tested period. This is a generic property of clustered data on the unit circle, and it fires at many arbitrary periods that have no proximity interpretation.

2.2 Empirical Demonstration: Period Sweep Dissociation

Table 1 demonstrates this concretely using the UNESCO context-validated dome/stupa corpus ($N = 83$). A period sweep tests both Rayleigh concentration and binomial proximity enrichment at the A++ precision threshold ($\pm 0.10 \times T$, geometric null $p_0 = 0.20$) at four candidate periods.

Table 1: Rayleigh concentration vs. A++ binomial proximity enrichment at four candidate periods, UNESCO dome corpus ($N = 83$, anchor 35.269°E). A++ threshold = $0.10 \times T$; geometric null $p_0 = 0.20$. Periods with Rayleigh significance show no proximity enrichment; the only period with genuine proximity enrichment (3.0) is Rayleigh-non-significant. The two statistics are anti-correlated.

Period	Rayleigh p	Sig	A++ hits	Expected	Binom. p	Sig
2.4	0.0023	**	21	16.6	0.1428	ns
3.0	0.1393	ns	26	16.6	0.0098	**
3.5	0.013	*	11	16.6	0.9588	ns
3.6	0.03	*	15	16.6	0.7113	ns

The anti-correlation is unambiguous. Rayleigh is significant at 2.4, 3.5, and 3.6, and non-significant at 3.0. The binomial proximity test is significant only at 3.0. A researcher applying the Rayleigh test would identify the wrong periods and miss the one with actual proximity structure.

The geometric explanation is direct. At 2.4, 3.5, and 3.6, one geographic cluster dominates the phase distribution, producing spuriously peaked R . At 3.0, the cluster structure spreads across multiple harmonics, lowering R while the binomial count of sites near any harmonic is elevated. The Rayleigh test detects geographic clustering unrelated to harmonic alignment; the binomial test detects the alignment directly.

3 The Geometric-Null Binomial Framework

3.1 The Geometric Null

For a grid of period T anchored at λ_0 , the deviation of site i from its nearest harmonic is

$$\delta_i = \min_{k \in \mathbb{Z}} |\lambda_i - (\lambda_0 + kT)|, \quad \text{folded to } [0, T/2].$$

A site falls within tier threshold d if $\delta_i \leq d$. Under a uniform longitude distribution, $\Pr(\delta_i \leq d) = 2d/T$ regardless of geographic clustering. Clustering is *neutral*: a dense cluster anywhere inflates both observed and expected hit counts proportionally, contributing to the binomial test only if the cluster falls asymmetrically near harmonic lines. Geographic concentration is therefore not a confound to be removed, but a source of statistical power that the test uses correctly.

3.2 Tier Classification

We define three proximity thresholds as round fractions of the $T/2$ half-period (1.5), calibrated to bracket the empirical natural proximity scales of the corpora under study: the OWTRAD Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%). Each threshold corresponds to a geometric null rate $p_0 = 2d/T$:

Tier	Threshold	Geometric null p_0
A++	$\delta \leq 0.06$ ($\lesssim 6.66$ km equatorial)	4%
A+	$\delta \leq 0.15$ ($\lesssim 16.6$ km)	10%
A	$\delta \leq 0.3$ ($\lesssim 33$ km)	20%

Tier labels are descriptive shorthand; the primary test is always the binomial.

3.3 Cross-Corpus Fisher Combination

For each corpus c at tier d , a one-sided binomial test yields p_c . Fisher’s combination statistic $\chi^2 = -2 \sum_c \ln p_c$, referred to $\chi^2(2C)$ for C corpora, tests whether the joint directional excess is consistent with chance across all corpora simultaneously. The geographic clusters of the three corpora are substantially non-overlapping, so the joint result cannot be attributed to a single shared geographic cluster. This three-corpus Fisher combination is the primary statistical result of the paper. Rayleigh statistics are retained for comparability with prior literature only.

4 Data

UNESCO dome/stupa subset ($N = 83$). The UNESCO World Heritage XML dataset ([UNESCO World Heritage Centre, 2024](#)) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text. Context validation reduces the raw match of $N = 90$ to $N = 83$, applying sentence-level gating to ambiguous terms while accepting unambiguous terms (stupa, tholos) without gating.

These results are conditional: the 3 / $\sim 35^\circ\text{E}$ frame was identified in this corpus first, so UNESCO results cannot independently confirm the frame.

Wikidata Q180987 stupa corpus ($N = 229$). All *instance-of: stupa* (Q180987) items in the Wikidata knowledge base, extracted without researcher involvement. Membership was fixed before any statistics were computed. The corpus is geographically dominated by the Java/Sumatra node ($\sim 105\text{--}115^\circ\text{E}$), reflecting the documented historical density of stupa construction around Borobudur (Miksic, 1990). This corpus carries no look-elsewhere conditioning.

OSM stupa layer ($N = 259$). OpenStreetMap features with tags `building=stupa`, `historic=stupa`, or `ruins=stupa`, assembled by crowd-sourced tagging independent of both UNESCO and Wikidata. Coverage correlates with mapping effort; the corpus is treated as corroborating evidence. Uneven contributor density is a coverage limitation, not a directional bias: it would produce an arbitrary anchor, not the same $\sim 35^\circ\text{E}$ line recovered by all three corpora.

Candidate frame. The reference anchor is Mount Gerizim (35.269°E), a prominent peak in the central Levantine highlands at the geographic junction between the Mediterranean coast and the interior corridor running toward Mesopotamia and beyond. It is not inscribed on the UNESCO World Heritage List. Its longitude falls within the candidate band identified in the data, and it serves here as a named label for that corridor-level phase rather than as a unique or privileged point. The anchor is band-invariant: every longitude in the $x.25^\circ\text{E}$ band produces the same proximity enrichment, and both Gerizim and Jerusalem (35.232°E , ≈ 4.1 km south) place the corridor in the global top 2.3% of 36,000 trial anchors. The data cannot distinguish Gerizim from any co-grid longitude within the same A+ tier.

5 Results

5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

All three corpora show consistent directional excess at every tier despite substantially different geographic footprints and assembly processes. The Fisher combination reaches conventional significance at all three tiers. The UNESCO A++ result is the strongest individual signal ($3.31\times$), reflecting the distributed character of that corpus; Wikidata and OSM A++ results are individually non-significant, reflecting cluster-level rather than individual-site precision. The Tier A Fisher combination ($p = 0.0$) is the most robust result.

5.2 Supplementary Rayleigh Phase Concentration

Table 3 reports Rayleigh statistics at $T = 3$ for comparability with prior literature. The full UNESCO corpus shows no Rayleigh periodicity ($R = 0.0224$, perm- $p = 0.603$, ns), confirming the signal is not a generic heritage-list property. Watson’s U^2 and Kuiper tests confirm the dome/stupa phase distribution differs detectably from the full UNESCO corpus ($p = 0.0036$, $p = 0.0013$), establishing monument-type specificity consistent with the geometric-null results.

Table 2: Cross-corpus geometric-null proximity test. One-sided binomial: observed harmonic-proximity hits vs. expected under $p_0 = 2d/T$. Fisher combination: $\chi^2(-2 \sum \ln p_i)$, $\chi^2(6)$, three corpora. UNESCO results are conditional; Wikidata and OSM results carry no such conditioning.

Tier	Corpus	N	Hits	Exp	Fold	p_{binom}	Sig
A++ ($p_0=4\%$)	UNESCO domes	83	11	3.3	$3.31\times$	0.0005	***
	Wikidata stupas	229	11	9.2	$1.2\times$	0.311	ns
	OSM stupas	259	12	10.4	$1.16\times$	0.3433	ns
	<i>Fisher</i> $\chi^2(6) = 19.785$, <i>combined</i> $p = 0.00302$						**
A+ ($p_0=10\%$)	UNESCO domes	83	16	8.3	$1.93\times$	0.0076	**
	Wikidata stupas	229	33	22.9	$1.44\times$	0.0212	*
	OSM stupas	259	33	25.9	$1.27\times$	0.089	
	<i>Fisher</i> $\chi^2(6) = 22.295$, <i>combined</i> $p = 0.00107$						**
A ($p_0=20\%$)	UNESCO domes	83	26	16.6	$1.57\times$	0.0098	**
	Wikidata stupas	229	70	45.8	$1.53\times$	0.0001	***
	OSM stupas	259	69	51.8	$1.33\times$	0.0059	**
	<i>Fisher</i> $\chi^2(6) = 37.999$, <i>combined</i> $p = 0.0$						***

Table 3: Supplementary Rayleigh phase concentration at $T = 3$, retained for comparability with prior literature. The primary evidence is Table 2.

Corpus	N	R	perm- p	Sig	Note
UNESCO full corpus	1011	0.0224	0.603	ns	Boundary condition
UNESCO dome/stupa	83	0.1571	0.1288	ns	Conditional
Wikidata Q180987	229	0.1669	0.0018	**	Independent
OSM stupa layer	259	0.1073	0.0355	*	Crowd-sourced
<i>Fisher</i> (<i>Wikidata</i> + <i>OSM</i>): $\chi^2(4) = 19.316$, $p = 0.0007$					

5.3 Negative Controls and Scope

The OWTRAD Silk Road route network ($N = 1674$ nodes) does not independently recover the monument anchor in the Normalized Phasor Consensus test ($K = 0.7429$, perm- $p = 0.46532$, ns). Using the 1946 edge midpoints as the proximity sample (one value per route segment, non-deduplicated), the binomial test is also non-significant at both tiers (A++: $p = 0.0908$, ns; A+: $p = 0.0679$, ns). The route network therefore fails both the directional and the per-segment proximity tests. By contrast, when route endpoints are pooled with multiplicity ($N = 3892$ node-in-edge occurrences, counting each city once per edge it connects), the binomial enrichment is significant (A++: $p = < 0.0001$, $z = 4.53$; A+: $p = 0.0002$). The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate of 9.7%, below the 10% geometric null. A Monte Carlo period sweep finds the dominant peak at 3 ($p < 0.001$); adjacent non-harmonic spacings (1.8, 2.1, 2.7, 3.3) are non-significant in both populations.

Null A (within-dome bootstrap, 100,000 trials) shows the UNESCO A++ count is fully reproducible from dome/stupa sites' own longitude distribution ($p = 0.5493$, ns): the enrichment relative to the full corpus is carried by where hemispherical monuments are located. Null C finds the broader same-footprint UNESCO background expects only

~ 5.1124 A++ hits versus the observed 11 ($p = 0.0129$ at the ± 5 window, surviving Bonferroni correction for $k = 3$ windows, $\alpha_{\text{adj}} = 0.0167$), establishing type-specificity within that geography.

5.4 Anchor Concordance

All three corpora recover the same $\sim 35\text{E}$ anchor. The Normalized Phasor Consensus statistic K — which compares normalized mean phasors at the candidate period, independent of tier counts — yields $K \geq 0.9695$ for all pairings of the three monument corpora (Table 4). The three-corpus combination reaches perm- $p = 0.01806$ with consensus anchor at 0.0 from 35E. Negative controls show K drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at $T = 3$ (100,000 permutation draws). High K with non-significant two-corpus perm- p reflects shared Eurasian geography. Negative controls (Rows 4–5) confirm K is not a generic Eurasian property.

Role	Corpora	K	perm- p	Note
Non-nested pair	Dome + Wikidata Stupa	0.9862	0.10479	Primary non-nested
Conservative	UNESCO full + Stupa	0.982	0.12243	Non-nested
Three-corpus	Dome + Stupa + OSM	0.9789	0.01806	All 3 monument corpora
Neg. control	UNESCO + Americas + Stupa	0.6723	0.29819	Geographic control
Contrast corpus	UNESCO + OWTRAD	0.7429	0.46532	Route network

6 Discussion

6.1 The Methods Contribution

The central finding is methodological. Table 1 shows empirically that the Rayleigh periodogram and the geometric-null binomial test are anti-correlated across periods in a geographically clustered monument corpus: Rayleigh significance appears at arbitrary periods carrying no proximity signal, and the only genuine proximity signal appears at a period where Rayleigh is non-significant.

This anti-correlation is a predictable consequence of what each test measures. The Rayleigh test detects unimodal phase concentration: the case where all sites share one preferred longitude modulo T . The binomial proximity test detects multimodal proximity enrichment: the case where sites fall near the full set of harmonics regardless of which specific harmonic they are near. These are different hypotheses. The one the archaeoastronomical literature has typically meant to test (“sites cluster near longitude harmonics”) is the latter, and using Rayleigh to test it introduces a systematic error.

The geometric-null binomial framework resolves this. The geometric null $p_0 = 2d/T$ is transparent, requires no distributional assumptions, and correctly treats geographic clustering as neutral. When multiple independently assembled corpora are available, Fisher combination of their binomial p-values gives a joint estimate whose significance is not inflated by within-corpus clustering, because the corpora’s geographic biases are non-overlapping. This framework is applicable to any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

6.2 Interpretation of the Empirical Illustration

The three-corpus demonstration finds consistent directional excess over the geometric null at the 3 / $\sim 35^\circ\text{E}$ candidate frame (Fisher $p = < 0.001$ at Tier A). The UNESCO corpus is the most geographically distributed result, with A-tier sites spanning the full Eurasian corridor. Null A shows its enrichment is reproducible from dome/stupa sites' own longitude geography ($p = 0.5493$, ns), confirming the signal is carried by where hemispherical monuments are located. Null C, however, shows that the broader same-footprint UNESCO background expects only ~ 5.1124 A++ hits versus the observed 11 ($p = 0.0129$), indicating partial type-specificity within that geography: the dome subset is more concentrated near harmonics than other monuments occupying the same longitude footprint. These two nulls point in opposite directions and should be read together: the enrichment is embedded in hemispherical monument geography (Null A), but that geography is itself non-randomly placed relative to the grid compared with the surrounding UNESCO background (Null C). The Wikidata result is substantially carried by the Java/Borobudur node, which is historically the dominant node of the stupa tradition (Miksic, 1990): the cluster's position on a 3 harmonic is the signal the geometric-null test is designed to detect. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, independently replicates the directional excess ($p = 0.0039$ at Tier A) and the same anchor ($K = 0.9695$ with Wikidata).

The present data neither confirm nor rule out any particular causal interpretation. The statistical pattern is consistent with several mechanisms that cannot be distinguished with the available evidence. Non-intentional explanations include geographic channeling by trade routes and river systems following the Eurasian corridor, cultural diffusion patterns concentrating stupa construction at historically dominant nodes (Sen, 2014), and selection effects in the UNESCO inscription process. An intentional placement interpretation would require pre-modern continental longitude determination at sub-degree precision, for which no documentary or archaeological evidence has been identified (Hunger & Steele, 2019; Neugebauer, 1975); but absence of evidence is not evidence of absence, and this paper does not attempt to adjudicate the question. The contribution of the present work is to establish what the correct statistical test is and what it finds when applied to three independent corpora — not to resolve the mechanism.

6.3 Limitations

The analysis is conditional on the stated candidate frame. An anchor-discovery correction places the Levantine corridor in the top 2.3% globally ($p = 0.0753$, †), a boundary condition on phase-free discovery claims. The type-specificity argument compares hemispherical monuments against the full UNESCO corpus, which includes non-architectural sites; a within-footprint comparison against non-hemispherical architectural monuments would provide a stronger test. Geocoding precision is a secondary concern: centroid displacement of 20–30 km for serial nominations is sufficient to cross the A+ tier boundary; the per-site audit (Tenenbaum, 2026) should be consulted for individual assignments.

7 Conclusion

The Rayleigh periodogram is structurally misaligned with the hypothesis it is being used to test in spatial monument studies. “Sites cluster near longitude harmonics” predicts

multimodal phase concentration across multiple harmonics, which produces low Rayleigh R . Using Rayleigh to test it fires on geographic clustering unrelated to harmonic alignment and misses genuine harmonic alignment. The period-sweep dissociation in Table 1 demonstrates this in a real monument corpus: Rayleigh significance and binomial proximity enrichment are anti-correlated across candidate periods.

The geometric-null binomial framework tests the correct hypothesis directly. Applied to three independently assembled hemispherical monument corpora at a $3 / \sim 35^\circ\text{E}$ candidate frame, the framework finds consistent directional excess (Fisher $p = < 0.001$ at Tier A). This demonstration is offered without causal claims: the mechanism gap is real, conditioning on a discovered frame is acknowledged, and non-intentional explanations receive equal standing. The contribution is the framework and the empirical demonstration of the Rayleigh-binomial dissociation.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>).

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Data Availability

All data, code, and audit files: Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>). UNESCO XML: UNESCO World Heritage Centre 2024.

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